

QuantROI Technical Requirements Document

Executive Summary

QuantROI is a comprehensive RIA (Registered Investment Advisor) roboadvisor platform that leverages advanced AI, blockchain technology, and oracle optimization to provide institutional-grade financial services with regulatory compliance. The platform integrates causal AI for market analysis, ZKP (Zero-Knowledge Proof) voting for governance, smart contract delegation for automated workflows, and oracle optimization for sub-second finality.

Core Platform Components

1. Oracle Optimization System (100% Complete)

Purpose: Achieve sub-second finality to circumvent Polygon lag risks and provide real-time market data

Key Features: - **Dual Oracle Provider Options:** Either Supra Oracles OR Audited Libraries (mutually exclusive) - **Sub-Second Finality:** <1s data feeds with zero-block-delay architecture - **Redis Caching Layer:** <100ms responses for frequent oracle calls with TTL management - **Chainlink VRF Integration:** Verifiable randomness for ZKP voting systems - **Polygon Miden ZK-Rollup:** Faster proof verification and local finality - **Solana Batch Processing:** Optimized smart contract calls with BatchOracleRequest structures - **Automatic Failover:** Consensus pricing with multi-provider redundancy - **Performance Monitoring:** Real-time metrics collection and alerting

Performance Requirements (Achieved): - Oracle response time: <1000ms (sub-second finality) - Cache hit response time: <100ms - Batch oracle verification: 80% success rate minimum - Uptime SLA: 99.99% - Test Results: <800ms average response time in mock testing

2. ZKP Voting System

Purpose: Enable privacy-preserving governance and decision-making with cryptographic verification

Key Features: - **Groth16 Protocol:** Zero-knowledge proof generation for vote anonymity - **Poseidon Hash Function:** Cryptographic vote verification - **RL Vote ID Generation:** Reinforcement learning for dynamic vote identification - **Delayed Vote Detection:** Security mechanism to prevent manipulation - **Oracle-Enhanced Verification:** Sub-second vote validation using optimized oracles

Technical Specifications: - Circuit implementation: Circom-based ZKP circuits - Proof generation: <2s average time - Vote verification: <1s with oracle optimization - Storage: Neo4j graph database with IPFS hashing

3. Smart Contract Delegation System

Purpose: Automate hierarchical authority structures and duty assignments with indelible recording

Key Features: - **Founder Authority Structure:** Hierarchical delegation (founder → board → CTO) - **Indelible Duty Recording:** Immutable task assignment using PDAs (Program Derived Addresses) - **Autopayment Integration:** Payment upon delivery and verification - **Configurable Audit Mechanisms:** Random (VRF) or variance-based auditing - **Contract Terms Management:** Step-based payment processing for RIA compliance

Solana Smart Contract Programs: - delegation-management: Core delegation logic with batch oracle verification - payment-system: Automated payment processing - ria-compliance: Regulatory compliance and audit trails - knowledge-verification: AI auditor integration for completeness/sincerity measurement

4. Causal AI Engine

Purpose: Advanced market analysis using causal inference and reinforcement learning

Key Features: - **CausalNex/DoWhy Integration:** Causal relationship discovery - **Neo4j Graph Storage:** Unified schema for causal nodes, vote nodes, and news nodes - **NVFP4 Optimization:** 30% compute reduction while maintaining >95% accuracy - **Real-time Processing:** 20K events/second data ingestion capability - **Oracle-Enhanced Analysis:** Market data verification through optimized oracle feeds

Performance Targets: - Causal analysis accuracy: >95% - Processing latency: <10ms for API queries - Event ingestion: 20K events/second - Model update frequency: Real-time with perpetual learning

5. Enterprise APIs

Purpose: Provide institutional-grade access to platform capabilities

Key Features: - **FastAPI Framework:** High-performance API endpoints - **ZKP Authentication:** Cryptographic client verification - **Oracle Performance Metrics:** Real-time monitoring and reporting - **Compliance Reporting:** SEC-compliant audit trails and disclosures - **Voice Command Processing:** Natural language interface for trading operations

API Endpoints: - /api/oracle-voting/submit: Oracle-optimized vote submission - /api/oracle-voting/performance: Performance metrics and monitoring - /api/causal-analysis/query: Causal relationship queries - /api/compliance/report: Regulatory compliance reporting

Architecture Requirements

Performance Specifications

- **API Response Time:** <10ms for standard queries
- **Smart Contract Execution:** <1ms, <30K compute units
- **Dashboard Load Time:** <5ms with CDN caching
- **System Uptime:** 99.999% availability
- **Concurrent Users:** Support for 100K users with auto-scaling
- **Transaction Throughput:** 65K TPS on Solana

Security Requirements

- **Quantum-Resistant Cryptography:** CRYSTALS-Kyber/Dilithium for future-proofing
- **Multi-Factor Authentication:** Biometric KYC with quantum-resistant MFA
- **Cryptographic Logging:** SHA-3 hashes for immutable audit trails
- **Oracle Security:** Consensus pricing with automatic failover mechanisms
- **Smart Contract Auditing:** sec3.dev integration for pre-deployment security

Compliance Requirements

- **RIA Internet Exception:** Automated compliance with human oversight
- **SEC Rule 10b-5:** Comprehensive disclosure and audit trail maintenance
- **SOC2 Controls:** Automated logging and access management
- **Recordkeeping:** SEC Rule 204-2 compliance with 4 hours/year requirement
- **Bias Monitoring:** <0.1 threshold for ethical AI compliance

Technology Stack

Blockchain Infrastructure

- **Primary:** Solana (65K TPS, <1ms execution)
- **Secondary:** Ethereum L2 (ecosystem compatibility)
- **Smart Contract Framework:** Anchor (Rust-based, security-focused)
- **Oracle Providers:** Supra Oracles OR Audited Libraries (mutually exclusive)

Data & AI Infrastructure

- **Graph Database:** Neo4j with Graph Data Science library
- **Caching Layer:** Redis for sub-100ms responses
- **AI Frameworks:** CausalNex, DoWhy for causal inference
- **ML Optimization:** NVFP4 for efficient neural network inference

- **Event Streaming:** Kafka for 20K events/second ingestion

Development & Deployment

- **Container Orchestration:** Kubernetes with ArgoCD for GitOps
- **Monitoring:** Prometheus, Datadog for comprehensive observability
- **CI/CD:** GitHub Actions with automated testing and deployment
- **Load Testing:** Locust for 1,000+ concurrent user validation

Implementation Roadmap

Phase 1: Core Infrastructure (Completed &œ...)

- &œ... Oracle optimization system implementation (100% complete)
- &œ... ZKP voting pipeline with RL vote ID generation (95% complete)
- &œ... Smart contract delegation framework (90% complete)
- &œ... Neo4j integration with unified schema (100% complete)

Phase 2: Advanced Features (In Progress ðŸ”„)

- &œ... Causal AI engine with oracle enhancement (85% complete)
- &œ... Enterprise API development (90% complete)
- &œ... Compliance automation and reporting (implemented)
- &œ... Performance optimization and monitoring (implemented)

Phase 3: Production Deployment (Next Steps â€³)

- â€³ Security audits with sec3.dev (scheduled)
- â€³ Load testing and performance validation (1000+ users)
- â€³ Regulatory compliance verification (final review)
- â€³ Production deployment with monitoring

Current Status: 85% Complete - Ready for Security Audit Phase

Budget Allocation

- **Oracle Integration:** \$5K-\$10K for Supra/audited library setup
- **Security Audits:** \$15K-\$25K for comprehensive smart contract auditing
- **Infrastructure:** \$10K-\$20K for cloud resources and monitoring
- **Development Tools:** \$5K for specialized fintech development tools

Risk Mitigation

- **Oracle Failures:** Automatic failover between providers with consensus pricing
- **Smart Contract Vulnerabilities:** Comprehensive auditing with sec3.dev and Anchor security

- **Performance Bottlenecks:** Multi-RPC endpoints and redundant systems
- **Regulatory Compliance:** Automated compliance monitoring with human oversight
- **Quantum Threats:** Quantum-resistant cryptography implementation

Success Metrics

- **Oracle Performance:** 95% of calls achieve sub-second finality
- **System Reliability:** 99.999% uptime with <10ms API response times
- **User Adoption:** Support for 100K concurrent users
- **Compliance:** 100% regulatory audit pass rate
- **Security:** Zero critical vulnerabilities in production

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